TERRAFORM COMMANDS CHEATSHEET

(A COMPLETE QUICK-REFERENCE GUIDE)

1. Terraform – Basics

Core commands to understand Terraform installation and usage.

- terraform version – Display installed Terraform version
- terraform help – Show general help
- terraform -help – List all available Terraform commands

2. Initialization

Prepare the working directory and backend.

- terraform init – Initialize Terraform working directory
- terraform init -upgrade – Upgrade providers & modules
- terraform init -reconfigure – Reconfigure backend settings
- terraform init -backend=false – Initialize without backend
- terraform init -migrate-state – Migrate state to a new backend

3. Validation & Formatting

Ensure configuration quality and consistency.

- terraform validate – Validate configuration files
- terraform validate -json – Validation output in JSON
- terraform fmt – Format Terraform files
- terraform fmt -recursive – Format files recursively
- terraform fmt -check – Check formatting without modifying
- terraform fmt -diff – Show formatting differences

4. Planning

Preview infrastructure changes before applying.

- terraform plan – Generate execution plan
- terraform plan -out=tfplan – Save plan to file
- terraform plan -var="key=value" – Pass variables at runtime
- terraform plan -var-file=vars.tfvars – Use variable file
- terraform plan -target=resource.name – Plan specific resource

- terraform plan -destroy – Preview destroy plan
- terraform plan -refresh=false – Skip state refresh

5. Apply

Provision or update infrastructure.

- terraform apply – Apply infrastructure changes
- terraform apply tfplan – Apply saved plan
- terraform apply -auto-approve – Skip confirmation
- terraform apply -var-file=vars.tfvars – Apply with variables
- terraform apply -target=resource.name – Apply specific resource

6. Destroy

Remove managed infrastructure safely.

- terraform destroy – Destroy all resources
- terraform destroy -auto-approve – Destroy without confirmation
- terraform destroy -target=resource.name – Destroy specific resource

7. State Management

Manage and troubleshoot Terraform state.

- terraform state list – List resources in state
- terraform state show <resource> – Show resource details
- terraform state pull – Download state from backend
- terraform state push – Upload state to backend
- terraform state mv <old> <new> – Move resource in state
- terraform state rm <resource> – Remove resource from state
- terraform state replace-provider <old> <new> – Replace provider

8. Outputs

Access output values from Terraform.

- terraform output – Show all outputs

- terraform output <name> – Show specific output
- terraform output -json – Output in JSON format
- terraform output -raw <name> – Display raw value

9. Variables & Console

Debug and test expressions interactively.

- terraform console – Open interactive Terraform console

10. Workspaces

Manage multiple environments (dev, stage, prod).

- terraform workspace list – List workspaces
- terraform workspace show – Show current workspace
- terraform workspace new <name> – Create workspace
- terraform workspace select <name> – Switch workspace
- terraform workspace delete <name> – Delete workspace

11. Providers & Modules

Handle providers and reusable modules.

- terraform providers – List required providers
- terraform providers mirror <dir> – Mirror providers locally
- terraform get – Download modules
- terraform get -update – Update modules

12. Import

Bring existing infrastructure under Terraform management.

- terraform import <resource> <id> – Import existing resource

13. Refresh & Show

Sync and inspect Terraform state.

- terraform refresh – Update state from real infrastructure

- terraform show – Display state or plan details
- terraform show -json – Output details in JSON

14. Graph

Visualize resource dependencies.

- terraform graph – Generate dependency graph

15. Debugging & Logs

Troubleshoot Terraform operations.

- TF_LOG=TRACE terraform apply – Enable trace-level logs
- TF_LOG=DEBUG terraform plan – Enable debug logs
- TF_LOG_PATH=terraform.log terraform apply – Save logs to file

16. Lock & Recovery

Handle state lock issues.

- terraform force-unlock <LOCK_ID> – Force unlock state

17. Advanced / Rare Commands

Used in special scenarios.

- terraform taint <resource> – Mark resource for recreation
- terraform untaint <resource> – Remove taint
- terraform test – Run Terraform tests *(new feature)*

18. Cleanup (Manual)

Local cleanup operations.

- rm -rf .terraform – Remove Terraform cache
- rm terraform.tfstate – Delete local state file

19. Terraform Workflow (One-Line Summary)

init → validate → plan → apply → state → destroy